

PVsyst - Simulation report

Grid-Connected System

Project: EC9180_PVsyst_Lab2.

Variant: New simulation variant

Sheds, single array

System power: 105 kWp

Kilinochchi - Sri Lanka

**PVsyst V8.1.5**

VC0, Simulation date:
14/08/26 16:24
with V8.1.5

Project summary**Geographical Site****Kilinochchi**

Sri Lanka

Situation

Latitude 9.3840 °(N)

Longitude 80.4087 °(E)

Altitude 29 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

Kilinochchi

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary**Grid-Connected System****Orientation #1****Fixed plane**

Tilt/Azimuth 9.5 / 0 °

System information**PV Array**

Nb. of modules

190 units

Pnom total

105 kWp

Sheds, single array**Near Shadings**

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

Inverters

Nb. of units

1 unit

Total power

100 kWac

Pnom ratio

1.05

Results summary

Simulation step Hourly

Produced Energy 168094 kWh/year Specific production 1609 kWh/kWp/year Perf. Ratio PR 83.67 %

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General parameters

Grid-Connected System

Sheds, single array

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 9.5 / 0 °

Sheds configuration

Nb. of sheds 5 units

Single array

Shading limit angle

Limit profile angle 22.4 °

Sizes

Sheds spacing 3.20 m

Sensitive width 2.29 m

GCR Shading 72.1 %

Top inactive band 0.02 m

Bottom inactive band 0.02 m

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic

Model Mono 550 Wp Twin half-cells bifacial
(Original PVsyst database)

Unit Nom. Power 550 Wp
Number of PV modules 190 units
Nominal (STC) 105 kWp
Modules 10 units x 19 In series

At operating cond. (50°C)

Pmpp 95.9 kWp
U mpp 718 V
I mpp 134 A

Total PV power

Nominal (STC) 105 kWp
Total 190 modules
Module area 491 m²
Cell area 453 m²

Inverter

Manufacturer Generic

Model 100 kWac inverter
(Original PVsyst database)

Unit Nom. Power 100 kWac
Number of inverters 2 * MPPT 50% 1 unit
Total power 100 kWac
Operating voltage 180-1000 V
Pnom ratio (DC:AC) 1.05
No power sharing between MPPTs

Total inverter power

Total power 100 kWac
Number of inverters 1 unit
Pnom ratio 1.05

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 88 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.38 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

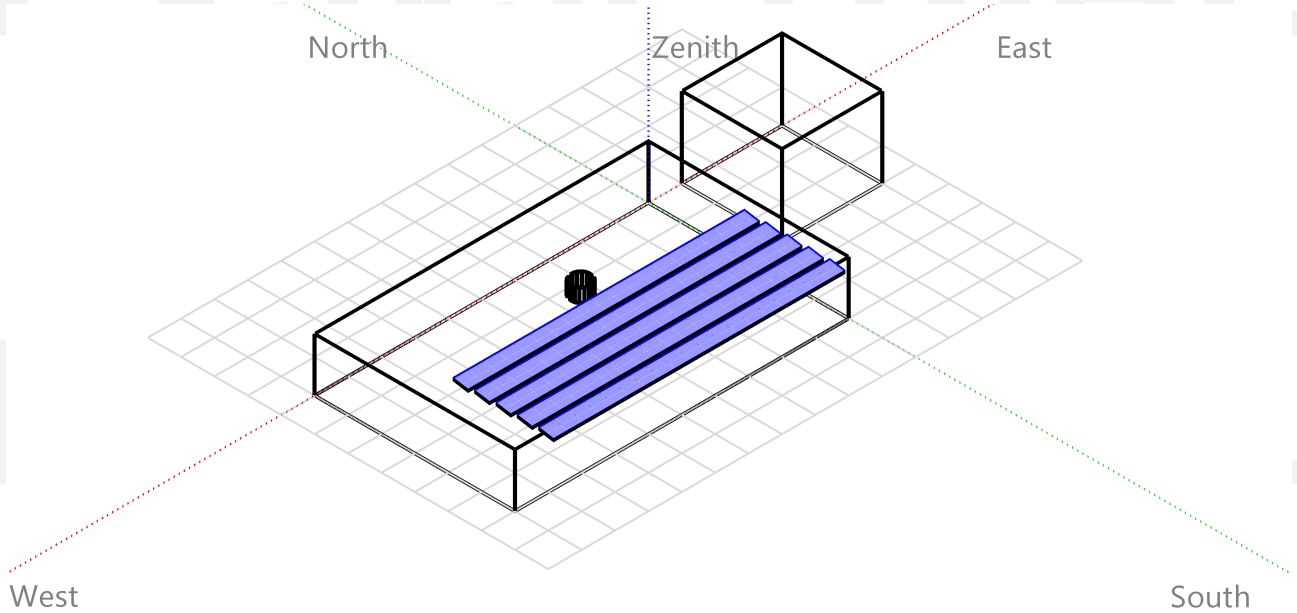


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Near shadings parameter

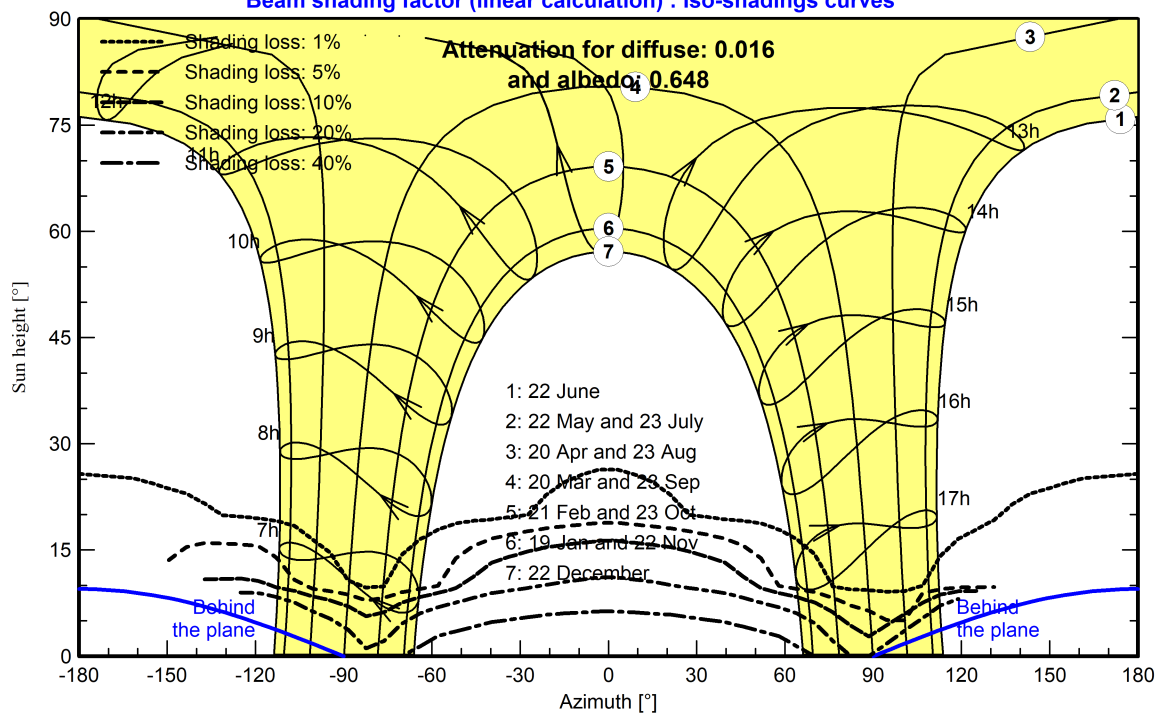
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 9.5°/ 0°

Beam shading factor (linear calculation) : Iso-shadings curves





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Main results

Simulation step

Hourly

System Production

Produced Energy

168094 kWh/year

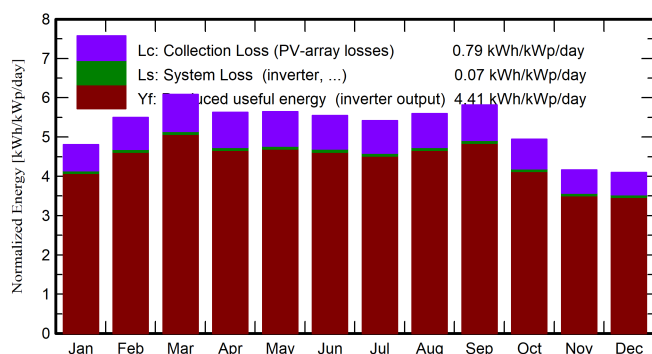
Specific production

1609 kWh/kWp/year

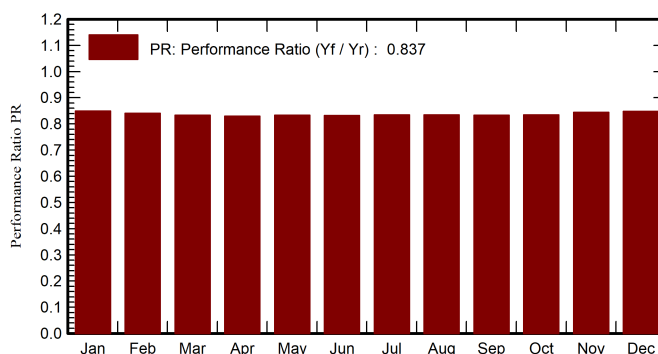
Perf. Ratio PR

83.67 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	138.4	67.23	25.56	149.0	144.8	13435	13213	0.849
February	146.2	60.86	26.51	154.0	150.1	13733	13519	0.840
March	184.8	78.57	28.57	188.6	183.9	16678	16430	0.834
April	171.9	81.15	30.10	168.9	164.1	14868	14639	0.830
May	182.2	89.05	30.32	174.9	169.9	15467	15232	0.833
June	176.0	80.75	30.28	166.5	161.5	14724	14488	0.833
July	176.3	86.42	29.86	167.9	162.8	14888	14646	0.835
August	177.9	83.32	29.34	173.4	168.6	15354	15111	0.834
September	173.1	77.08	28.88	174.4	169.8	15428	15196	0.834
October	147.7	71.46	27.95	153.2	148.9	13595	13370	0.835
November	117.8	67.10	26.72	124.8	120.9	11205	11005	0.844
December	118.1	64.60	25.79	126.9	122.9	11453	11245	0.848
Year	1910.5	907.58	28.33	1922.5	1868.1	170830	168094	0.837

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

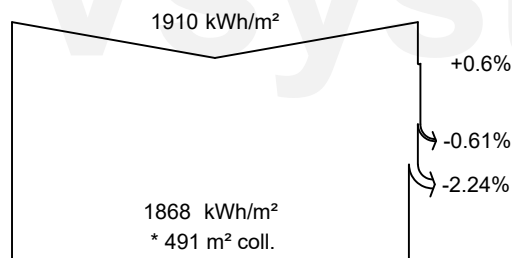
PR Performance Ratio



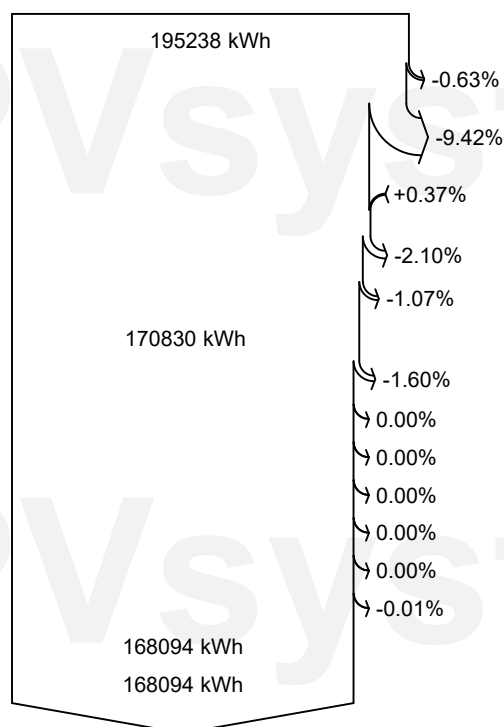
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Loss diagram



efficiency at STC = 21.29%



Global horizontal irradiation

Global incident in coll. plane

Near Shadings: irradiance loss

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

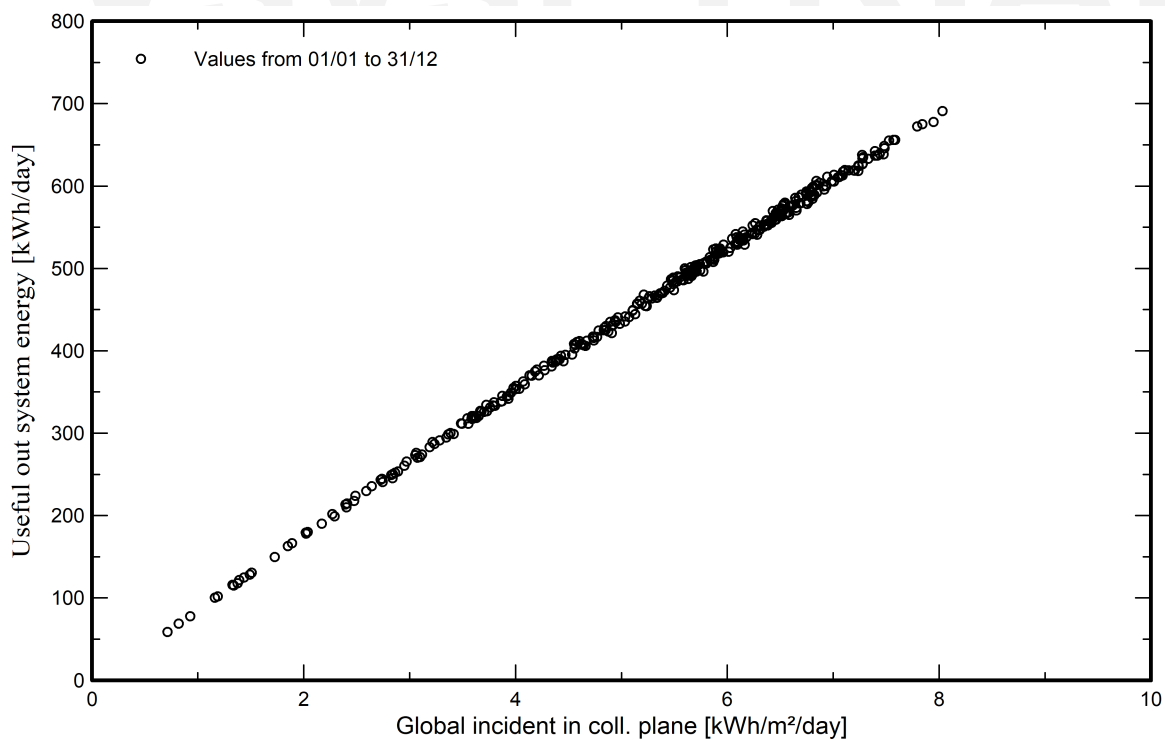
Energy injected into grid



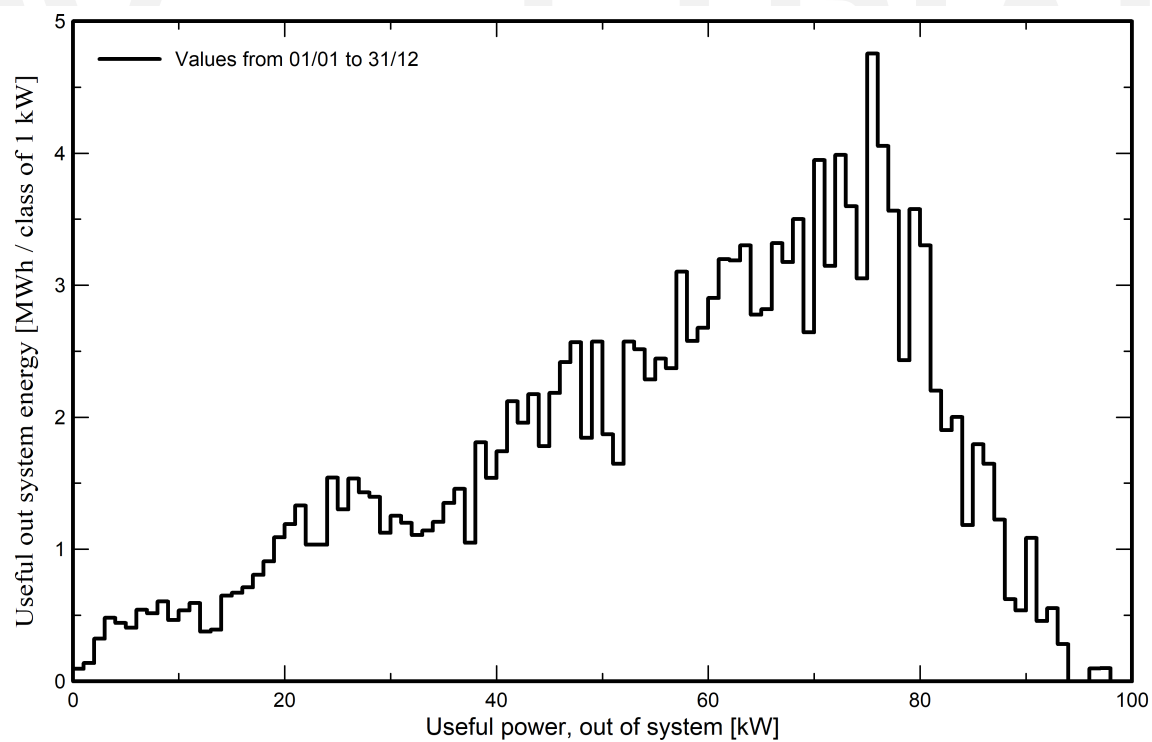
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

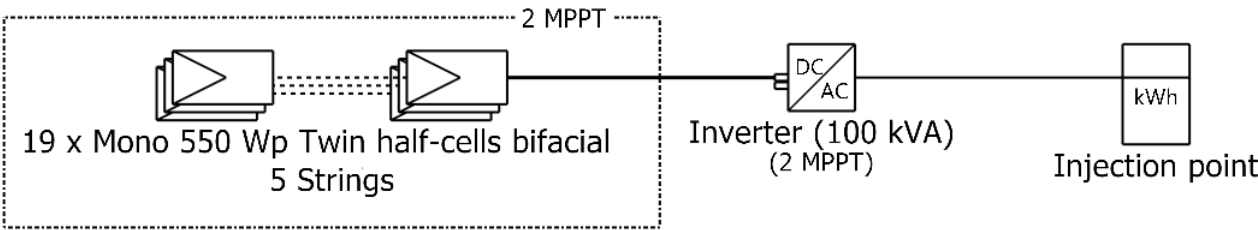




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Single-line diagram



PV module	Mono 550 Wp Twin half-cells bifacial
Inverter	100 kWac inverter
String	19 x Mono 550 Wp Twin half-cells bifacial

EC9180_PVsyst_Lab2

VC0 : New simulation variant

14/08/26